

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 01

Climate Introduction

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	1.0 - Climate Introduction (21 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 16 PDF pages)
Method	Question wording and answer-content matched to specific lecture slides. A link is counted only when the deck gives a usable definition, list, comparison element or decision framework; mere syllabus mention is not counted.

RESULT

14 past-question links were found across seven of the eight supplied papers, representing 175 mapped marks. The strongest directly covered cluster is Climate - Context - Comfort (2016, 2017, 2018 and 2024). Designer-responsibility and passive/active-control questions recur often, but this lecture provides only a foundation; their complete answers belong to later lectures.

Prepared as part of a cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 01 is a foundational concept lecture. It directly supports the recurring Climate-Context-Comfort family and introduces the design-with-environment / mitigation logic. Historical examination section labels vary, so the mapping below follows subject matter and the current Section A classification rather than old printed A/B labels.

21	14	3	2	9	175
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Classification	Meaning
Exact	The requested definition/list/comparison appears essentially as asked and can be answered directly from the cited slides.
Direct	Most of the answer framework is present; the student must explain, synthesize or apply the slide content.
Partial	Only a supporting component is present; another lecture/source is required for a complete answer.

Section-label note

The source deck cover says “Climate & Design (B)”. It is treated here as Section A - Lecture 01 because that is how the lecture was supplied and organized for the current course preparation. Historical exam questions are mapped by content, not by their old printed section label.

Slide-numbering note

The mapping uses PDF page/slide numbers with the title page as Slide 1. In this deck, the visible numbers from Slide 2 onward match the PDF page count.

Highest-priority slide blocks

Priority	Slides	Topic	Reason
1	19-20	Climate - Context - Comfort + design considerations	Directly supports the four-paper CCC cluster and most of the conceptual short-answer content.
2	16-18	Mitigation + design with environment + why forms differ	Supports the 2018 “accept/reject/mitigate” question and climate-study/application questions.
3	4-12	Man-made environment; shelter and comfort	Directly supports the 2016 short note on Man-made Environment and gives intuitive examples.
4	13-15	Environmental cost / consequence of development	Contextual support for mitigation and “why climate study matters”; not a standalone past-question answer.
5	21	Course-content overview	Useful as a navigation slide only; do not treat the syllabus list as complete answer content.

2. Detailed Year-wise Question Mapping

Question wording has been lightly standardized for readability without changing the assessed meaning. “QB p.” is the page in the uploaded 2016-2024 regular question bank.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Sec A Q1(a)	10	1	Figure out the importance of climate study for a BECM graduate.	Direct	17-20	Slides 17-20 establish design with environment, climatic design as environmental design, course objectives and climate-linked design considerations. A profession-specific BECM checklist is not stated.
2016	Sec B Q5(a)	15	1	Write short notes on: (i) Context, (ii) Comfort and (iii) Man-made Environment.	Exact	4-5, 19-20	Man-made environment is introduced through shelter/comfort; Slides 19-20 directly define Context and Comfort and summarize the three concepts.
2017	Sec A Q1(a)	15	2	Identify the areas where you can utilize your learning from the course “Climate and Architectural Design” as a BECM professional.	Partial	17-21	The deck states the environmental-design purpose, objectives and course scope. It does not give a profession-specific list of BECM application areas.
2017	Sec B Q5(a)	10	2	Explain “Climate” and “Context” from the point of view of a built-form designer.	Exact	19-20	Slides 19-20 give the core definitions, focus, key factors and design considerations for both Climate and Context.
2017	Sec B Q6(c)	20	2	What are the responsibilities of the built-form designers? What kind of complexities do they have to face in Climate Design?	Partial	18-20	Slides 18-20 support climate-responsive design decisions, but do not develop the formal responsibility/complexity lists. Full treatment belongs to Lecture 04.
2018	Sec A Q1(a)	10	3	What differences exist among Climate, Context and Comfort?	Exact	19-20	The comparison table on Slide 20 directly contrasts definition, focus, key factors and design considerations for all three terms.
2018	Sec A Q1(b)	15	3	Defend your position about Climate and Built Environment: should we “accept the fate”, “reject the development” or mitigate?	Partial	13-17	Slide 16 presents the exact dilemma and Slide 17 points to “Design with Environment”. The detailed argumentative justification is not developed in this deck.
2018	Sec B Q8(a)	10	4	Distinguish between passive and active means of thermal control. Which one do you prefer for the best indoor thermal condition?	Partial	19-20	Slide 19 names mechanical, structural and ventilation control; Slide 20 mentions passive design strategies. The active-vs-passive comparison and preference logic are not fully taught here.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2020-2022

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2020	Sec B Q4(a)	15	5	Differentiate active and passive thermal control techniques. Which one do you prefer as a BECM professional?	Partial	19-20	Same limitation as 2018: the deck introduces control categories and passive strategies, but not a full comparative framework.
2020	Sec B Q5(a)	15	5	What are the responsibilities of the built-form designers? What kinds of complexities do they have to deal with in climate design?	Partial	18-20	Useful design-consideration framework is present, but the formal responsibilities and climate-design complexities are absent.
2021	Sec B Q5(c)	10	6	Discuss the designers' responsibilities in ensuring thermal comfort in built environment.	Partial	19-20	Comfort and related design considerations are named, but thermal-comfort responsibilities are not developed. Use Lecture 04 plus thermal-comfort lectures.
2022	Sec B Q5(a)	10	8	Discuss the designer's responsibilities in ensuring thermal comfort in building environment. Mention design tools for moderating microclimate.	Partial	19-20	The deck gives broad control/design categories only. It does not provide the detailed responsibility or microclimate-tool lists required for a complete answer.

2023 diagnostic

No 2023 question was counted as a meaningful Lecture-01 answer-content match. Several 2023 questions concern thermal comfort, site climate, microclimate and climate-friendly construction, but Lecture 01 only mentions or frames those themes; the usable answer content is in later lectures.

2. Detailed Year-wise Question Mapping (continued)

Examination 2024

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2024	Sec B Q5(a)	10	14	Explain the differences between climate, context and comfort in architectural design. Discuss how each factor influences design decisions.	Direct	19-20	Slide 20 is almost a ready-made answer: definitions, focus, key factors and design considerations are all tabulated. The student only needs to explain and connect them.
2024	Sec B Q6(b)	10	14	Discuss the designer's responsibilities in ensuring thermal comfort in built environment and state the domains in which a designer should have knowledge.	Partial	19-20	The lecture supplies a general comfort/design framework but not the required responsibility and essential-knowledge domains. Lecture 04 is the direct source.

Current-paper signal

The 2024 paper returned directly to the Lecture-01 core: Climate, Context and Comfort were asked together for 10 marks. This reinforces Slides 19-20 as the single highest-return block in this lecture.

3. Repeated Question Patterns

A repeated cluster means the wording is identical or the tested decision/framework is substantially the same. Frequency alone is not enough: the “coverage” column shows whether Lecture 01 can actually answer it.

#	Pattern	Years / questions	Slides	Coverage / exam insight
1	Climate - Context - Comfort definitions / differences	2016 Q5(a); 2017 Q5(a); 2018 Q1(a); 2024 Q5(a)	4-5, 19-20	4-paper cluster and the best directly covered family. Master Slide 20 as a comparison table.
2	Designer responsibility / climate-design application	2017 Q6(c); 2020 Q5(a); 2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	18-20	5 appearances, but only partial coverage here. Lecture 04 carries the formal responsibility, essential-knowledge and complexity lists.
3	Passive vs active thermal-control decision	2018 Q8(a); 2020 Q4(a)	19-20	Repeated 10-15 mark comparison. Lecture 01 introduces controls but does not teach the full comparison.
4	Importance / professional use of climate study	2016 Q1(a); 2017 Q1(a)	17-21	Use design-with-environment, course objectives and scope. The profession-specific application list needs synthesis.
5	Accept fate / reject development / mitigate	2018 Q1(b)	13-17	The exact dilemma is on Slide 16; Slide 17 provides the answer direction (“Design with Environment”), but the argument must be expanded elsewhere.

Quick recognition rules

Question clue	Go first to	What Lecture 01 can do
“Context / Comfort / Man-made Environment”	Slides 4-5, 19-20	Direct definitions + shelter/comfort logic.
“Differences among Climate, Context, Comfort”	Slides 19-20	Use Slide 20 as the answer skeleton: definition -> focus -> key factors -> design considerations.
“Importance / use of climate study”	Slides 17-21	Explain design with environment, objectives and course scope; add BECM-specific application only if taught elsewhere.
“Accept fate / reject development / mitigate”	Slides 13-17	State environmental cost, reject the two extremes, choose mitigation / design with environment.
“Designer responsibilities / complexities”	Slides 18-20, then Lecture 04	Lecture 01 is only a bridge; do not stop here.
“Passive vs active thermal control”	Slides 19-20, then Lecture 04	Controls are introduced, not fully compared.

4. Slide-to-Question Reverse Index

Use this page while studying the lecture: after finishing a slide block, immediately practice the listed past questions. “Support” means the block helps the answer but is not sufficient by itself.

Slides	Topic	Past questions	Study note
2-3	Environment - visual introduction	2016 Q5(a) (support)	Concept introduction only; no formal environment definition is given.
4-5	Man-made environment; shelter; comfort	2016 Q5(a)	High-value for the “Man-made Environment” short note.
6-12	Examples of human abode and comfort	2016 Q5(a) (support)	Use as intuitive examples; do not reproduce images as theory.
13-15	“In what cost?” - environmental consequence	2016 Q1(a); 2018 Q1(b) (support)	Builds the rationale for climate study and mitigation.
16-17	Accept / reject / mitigate; Design with Environment	2016 Q1(a); 2018 Q1(b)	Core decision block. Slide 17 states that climatic design is part of environmental design.
18	Why forms, materials and openings differ	2016 Q1(a); 2017 Q1(a); 2017 Q6(c); 2020 Q5(a) (support)	Useful transition from climate conditions to actual building decisions.
19	Climate / Context / Comfort + objectives + controls	2016 Q5(a); 2017 Q5(a); 2018 Q1(a), Q8(a); 2020 Q4(a), Q5(a); 2021 Q5(c); 2022 Q5(a); 2024 Q5(a), Q6(b)	Highest-density slide; direct for CCC, partial for responsibility/control questions.
20	Detailed CCC comparison table	2016 Q5(a); 2017 Q5(a); 2018 Q1(a); 2024 Q5(a)	Single most exam-efficient slide in Lecture 01.
21	Course-content overview	2017 Q1(a) (support)	Navigation only. The listed topics are not taught in detail on this slide.

5. Coverage Gaps and Diagnostic Exclusions

The value of a mapping report is not only showing what a lecture covers, but also preventing you from forcing an incomplete lecture into a question it cannot answer.

Gap / not fully taught in Lecture 01	Affected past questions	Action
Designer responsibilities, essential knowledge and climate-design complexities	2017 Q6(c); 2020 Q5(a); 2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	Use Lecture 04 - Climate & Built Environment. Lecture 01 only supplies the broad design framework.
Passive vs active thermal-control comparison	2018 Q8(a); 2020 Q4(a)	Use Lecture 04 / dedicated thermal-control material. Lecture 01 names control types but does not build the comparison.
Climate vs Environment as a formal comparison	2017 Q6(a)	The deck shows “Environment” and “Man-made Environment” but does not define Environment analytically. Use Lecture 02 / class note.
Detailed thermal comfort, physiology and heat exchange	2016 Q6; 2018 Q7; 2021-2024 comfort questions	Slide 20 only lists thermal comfort under Comfort. Use the thermal-comfort / climatic-comfort lectures.
Site-climate factors and large-site design process	2020 Q3; 2021-2024 site/urban questions	“Context” is introduced here, but no site-analysis procedure is taught. Use the regional/site-climate lecture.
Green building, tube house, rat-trap bond, contemporary materials	2016-2024 climate-friendly construction questions	Slide 21 only names these in the course-content list. Use Lecture 10; do not treat a syllabus mention as answer content.
Ventilation calculations / air-space / infiltration	2021-2024 numerical questions	Slide 19 only names “Ventilation system”. Use Lecture 11 for definitions, tables and numericals.

Mapping discipline

Not counted: 2023 questions and other later-topic questions were not linked merely because Slide 21 lists their syllabus title. A question link is reserved for slides that provide usable answer content, not just a topic name.

6. Recommended Study Plan

Study this lecture by exam return, not by slide order. The goal is to master the high-value conceptual framework first, then use the remaining slides to strengthen reasoning and examples.

Step	Study block	Slides	Practice immediately
1	Master Climate - Context - Comfort comparison	19-20	2016 Q5(a); 2017 Q5(a); 2018 Q1(a); 2024 Q5(a)
2	Mitigation and "Design with Environment" logic	13-18	2016 Q1(a); 2018 Q1(b)
3	Man-made environment: shelter -> comfort	4-12	2016 Q5(a)
4	Course purpose / professional-use framework	17-21	2017 Q1(a)
5	Use Lecture 01 only as a bridge for responsibility/control questions	18-20	2017 Q6(c); 2018 Q8(a); 2020 Q4(a), Q5(a); 2021 Q5(c); 2022 Q5(a); 2024 Q6(b) - then continue to Lecture 04

Exam-ready minimum from Lecture 01

- Climate: given / long-term environmental condition - especially temperature, humidity, precipitation, wind and solar radiation.
- Context: landform, geography and the physical/cultural setting within which the building must fit.
- Comfort: desirable occupant condition - thermal, visual, acoustic and air-quality satisfaction.
- Man-made environment: human modification of the environment to obtain shelter and comfort.
- Mitigation position: neither accept environmental stress passively nor reject development; design with the environment to reduce impact.
- Climatic control framework: mechanical control, structural control and ventilation system.
- High-return answer structure: Definition -> key factors -> design consideration -> one practical example -> conclusion.

Final recommendation

If time is short, memorize Slides 19-20 first. They answer the most repeated direct cluster. Then study Slides 16-18 for the mitigation / design-with-environment argument. Treat responsibility, passive-active control, thermal comfort, site climate and climate-friendly construction as bridge topics only in Lecture 01 and move to their dedicated lectures for full answers.